Tank 90 — Complete ICP-MS Assessment & Dosing Plan

Sample MSR235399 | 29.04.2026 | 90 Gallon | LPS Coral Load | ESV B-Ionic 2-Part | RM ICP-MS Criteria

Report: **May 03, 2026** | Reef Moonshiners Handbook Rev.5

Key differences from Tank 150: Tank 90 is LPS-only with a lighter coral load and no macroalgae, so daily element consumption rates are significantly lower — doses are scaled accordingly. ESV B-Ionic 2-part replaces the BRS powder system, and is specifically designed for equal-volume dosing of both components. Alkalinity is already in target (8.0 dKH) — no Alk correction needed. pH probe has non-linear drift and needs cleaning or replacement before pH readings can be trusted.

Section 1 — Full ICP-MS Assessment

Element	Value	Unit	Target	Status	Assessment
Calcium	462	mg/L	420–440	**ABOVE**	Elevated — let settle naturally. Stop Ca pump until Ca reaches 435. Then restart at 10 ml/day (equal to Alk pump). Do NOT dose Ca while elevated.
Magnesium	1339	mg/L	1300–1400	**IN TARGET**	In range. ESV B-Ionic maintains Mg naturally. No correction needed.
Potassium	404	mg/L	400–420	**IN TARGET**	In target — previous correction worked perfectly.
Bromide	90	mg/L	80–90	**IN TARGET**	At upper end of target. No action needed.
Boron	6.1	mg/L	5.9–6.5	**IN TARGET**	In range. No action.
Alkalinity	8.0	dKH	7.8–8.5	**IN TARGET**	Perfect. Maintain Alk pump at 10 ml/day ESV Component 1.
Salinity	38.4	PSU	34–35	**ABOVE**	ICP salinity reading is based on Na/Cl ratio — verify with calibrated hydrometer. Same pattern as Tank 150. If confirmed high, dilute slowly with RODI (0.5–1.0 PSU/day max).
Strontium	4.9	mg/L	9–12	**ACTION**	Low — correction needed urgently. 5.36 ml/day x 6 days (RM Strontium UHP).
Barium	5.12	µg/L	10–20	**ACTION**	Low — Barium sponge effect active. Correction to 30 µg/L: 21.19 ml/day x 4 days.
Fluoride	0.65	mg/L	1.0–2.0	**ACTION**	Below target. 32.18 ml/day x 9 days (RM Fluoride) into sump.
Zinc	0.13	µg/L	4.0–6.0	**ACTION**	Depleted. 0.55 ml/day x 3 days (RM Zinc). Treat as monthly correction routine.
Caesium	1.43	µg/L	4–6	**ACTION**	Low — linked to RTN risk in LPS. 0.61 ml/day x 2 days (RM Caesium).
Molybdenum	14.5	µg/L	13–17	**IN TARGET**	In range. Small correction still advised: 1.70 ml x 1 day.
Nickel	1.52	µg/L	2.5	**ACTION**	Below target. 1.67 ml/day x 2 days correction (RM Nickel). Then maintain at 1.6 ml/day.
Iodine	50.8	µg/L	75–95	**ACTION**	Low despite 4 drops/day dosing. Increase to 6–8 drops/day Seachem Reef Iodide. Critical for LPS — Hammers and Torches respond strongly to Iodine.
Iron	0.05	µg/L	0.70–1.0	**ACTION**	Truly depleted. Start daily dose immediately. Increase from 1.0 to 2.0 ml/day Classic.
Cobalt	0.10	µg/L	0.40–0.60	**ACTION**	Well below target. Increase from 0.4 to 1.0–1.2 ml/day Classic.

Element	Value	Unit	Target	Status	Assessment
Copper	0.08	µg/L	0.40–0.60	**ACTION**	Depleted. Current 2.0 ml/day NANO is insufficient. Increase to 6–8 ml/day NANO.
Chromium	0.41	µg/L	0.35–0.55	**IN TARGET**	Perfectly calibrated at 0.4 ml/day Classic. Maintain — do not change.
Manganese	0.10	µg/L	0.40–0.60	**MONITOR**	ICP-MS detection limit issue — may be higher than shown. Maintain 2.0 ml/day Classic. Next ICP will calibrate.
Selenium	0.272	µg/L	0.35–0.55	**ACTION**	Not previously dosed — below target. Start at 0.07 ml/day Classic (1x standard).
Vanadium	0.33	µg/L	1.0–2.0	**ACTION**	Below target at 1 drop/day. Increase to 3 drops/day. LPS especially Hammers and Torches respond visibly to Vanadium.
Rubidium	264.2	µg/L	250–310	**IN TARGET**	In target. No action needed. Beneficial for LPS, soft corals.
Lithium	412	µg/L	50–500	**IN TARGET**	In range but monitor — ESV Ca component contains Lithium as impurity. Watch for upward trend between ICPs.
Aluminum	2.3	µg/L	0–5	**IN TARGET**	Low — good.
Silicon	179	µg/L	50–350	**IN TARGET**	In range.
Nitrate	4.86	mg/L	0–50	**IN TARGET**	Normal range.
Phosphate	0.061	mg/L	0.06–0.09	**IN TARGET**	Ideal range for LPS reef. No action.

Section 2 — Calcium, Alkalinity & ESV B-Ionic 2-Part

Parameter	Current	Target	Status	Action
Alkalinity	8.0 dKH	7.8–8.5	✓ IN TARGET	Maintain Alk pump at 10 ml/day ESV Component 1. No change needed.
Calcium	462 mg/L	420–440	ABOVE	Stop Ca pump. Ca falls ~4.6 mg/L/day naturally. Reaches 435 in ~6 days. Then restart at 10 ml/day (equal to Alk).

ESV B-Ionic is designed for equal volumes of each component. Running Ca at 15 ml/day vs Alk at 10 ml/day (1.5:1 ratio) directly explains the elevated Calcium. Once Ca is in range, run both pumps at equal volumes: 10 ml/day each. This is the correct long-term maintenance for this tank's consumption rate (0.23 dKH/day Alk — consistent with LPS-only lighter coral load). ESV also contains Mg in the Ca component — no separate Mg supplementation needed.

Section 3 — Correction Dosing Schedule

Dose corrections into sump/overflow. Allow 1 minute between elements. Never pre-mix.

Element	Current	Target	ml/Day	Days	Product	Notes
Strontium	4.9 mg/L	10 mg/L	**5.36**	**6**	RM Strontium UHP	Low — immediate correction needed. Spread over 6 days.
Barium	5.12 µg/L	30 µg/L	**21.19**	**4**	RM Barium	Sponge effect active — targeting 30 µg/L. 4-day correction.
Fluoride	0.65 mg/L	1.5 mg/L	**32.18**	**9**	RM Fluoride	Below target. Largest correction volume — 9-day spread.
Nickel	1.52 µg/L	2.5 µg/L	**1.67**	**2**	RM Nickel	Correction then maintain at 1.6 ml/day.
Zinc	0.13 µg/L	5 µg/L	**0.55**	**3**	RM Zinc	Depleted. Add to monthly correction routine going forward.
Caesium	1.43 µg/L	5 µg/L	**0.61**	**2**	RM Caesium	Low Caesium linked to RTN risk in LPS.
Molybdenum	14.5 µg/L	15 µg/L	**1.70**	**1**	RM Molybdenum	Just below target. Single-day correction.
Magnesium	1339 mg/L	1350 mg/L	**90.04**	**1**	BRS Mag-Mix (400g/L RODI)	Slight deficiency. Single-day correction.

Section 4 — Daily Dosing Plan

Dose dailies into display tank. Individual syringes — never pre-mix. Iodine and Vanadium: cobalt-blue dropper bottles. LPS doses are intentionally lower than Tank 150 — scaled for lighter coral load.

Element	Previous	New Dose	Notes
Iron (Classic)	1.0 ml	**2.0 ml/day**	Truly depleted at 0.05 µg/L. Increase immediately. LPS consumption lower than Tank 150 — 2.0 ml is a conservative increase. Calibrate from next ICP.
Copper (NANO)	2.0 ml	**6–8 ml/day**	Depleted at 0.08 µg/L. 2.0 ml NANO delivers only 0.011 µg/L/day — far too low. Increase to 6–8 ml/day NANO.
Cobalt (Classic)	0.4 ml	**1.0–1.2 ml/day**	Below target at 0.10 µg/L. 0.4 ml delivers 0.119 µg/L/day but all being consumed. Increase to 1.0–1.2 ml/day Classic.
Iodine	4 drops	**6–8 drops/day**	50.8 µg/L — below 75–95 target despite dosing. Seachem Reef Iodide ONLY. Cobalt-blue dropper bottle. LPS especially Hammers and Torches are very Iodine-sensitive.

Element	Previous	New Dose	Notes
Vanadium	1 drop	**3 drops/day**	0.33 µg/L — well below 1.0–2.0 target. LPS corals particularly Torches and Hammers respond visibly to Vanadium. Cobalt-blue dropper bottle.
Selenium (Classic)	Not dosed	**0.07 ml/day**	0.272 µg/L — below 0.35–0.55 target. Not previously dosed. Start at 1x standard (0.07 ml/day Classic). Very small volume — use a dropper bottle or precision syringe.
Manganese (Classic)	2.0 ml	**2.0 ml/day**	Maintain current dose. ICP-MS detection limit issue — 0.10 µg/L shown but likely higher. LPS tank won't need Tank 150 volumes. Continue and calibrate from next ICP.
Chromium (Classic)	0.4 ml	**0.4 ml/day ✓**	Perfectly calibrated — 0.41 µg/L in target (0.35–0.55). This is the benchmark dose for Tank 90. Do not change.
Nickel (Classic)	1.45 ml	**1.67 ml/day x2 then 1.6 ml/day**	Correction: 1.67 ml/day for 2 days. Then maintenance at 1.6 ml/day Classic. Previous 1.45 ml/day maintained ~1.52 µg/L — increase to 1.6 ml/day to hold at 2.5 target.

Section 5 — pH Probe Status

Reading	Reported	Est. True	Assessment
Morning low	8.39	~7.90	Borderline — slightly low
Average	8.53	~8.05	✓ Normal
Afternoon high	8.66	~8.20	✓ Normal healthy range

Pre-calibration readings: pH 7.00 buffer read 7.66 (+0.66), pH 10.00 buffer read 10.30 (+0.30). Non-linear drift confirmed — electrode slope 88.0% (below 90% replacement threshold). True pH is likely 7.90–8.20 — borderline normal but the low morning value warrants attention once a reliable probe is in place. Action required: (1) Soak junction in warm RODI 30 min, then KCl 1–2 hours. (2) Re-test in pH 7.00 buffer — if still above 7.40, replace probe. (3) Compatible Neptune Apex replacements: Neptune OEM (~$65), Pinpoint (~$55), Milwaukee MA911B/2 (~$30).

Section 6 — Next Steps & Monitoring

Immediately	Stop Ca pump (ESV Component 2). Ca will fall to 435 mg/L in ~6 days. Verify salinity with calibrated hydrometer — ICP salinity reading is indicative only.
pH probe — today	Soak reference junction in warm RODI 30 min then KCl 1–2 hours. Re-test in pH 7.00 buffer. If still above 7.40 or slope below 90% — replace probe. Once reliable probe is in place, monitor morning pH. If true morning low is consistently below 7.90 on the new probe, consider refugium reverse photoperiod to smooth the pH curve.
Start corrections — Day 1	Strontium (5.36 ml/day x 6), Barium (21.19 ml/day x 4), Fluoride (32.18 ml/day x 9), Nickel correction (1.67 ml/day x 2), Zinc (0.55 ml/day x 3), Caesium (0.61 ml/day x 2), Molybdenum (1.70 ml x 1), Magnesium (90.04 ml x 1). All into sump/overflow, 1 minute between each.
Update daily doses	Iron: 2.0 ml/day. Copper NANO: 6–8 ml/day. Cobalt: 1.0–1.2 ml/day. Iodine: 6–8 drops/day. Vanadium: 3 drops/day. Selenium: start 0.07 ml/day (new). Chromium: maintain 0.4 ml/day (perfect). Manganese: maintain 2.0 ml/day. Nickel: 1.6 ml/day after 2-day correction.
~6 days	Ca should be near 435 mg/L. Restart Ca pump at 10 ml/day (equal to Alk pump). Test Ca to confirm before restarting.

Send next ICP in 2 weeks	Take sample before daily dosing. Key calibration points: Iron, Copper, Cobalt consumption rates are unknown for this tank — next ICP gives the first real calibration data. Also confirms Iodine trajectory, Strontium and Barium hold, and whether Ca/Alk balance has stabilised at equal volumes.
Monthly	Recalibrate pH probe (new probe once replaced) with fresh sealed buffers. Check Ca and Alk with home test kits. Zinc should be treated as a monthly correction element for this tank.

Tank 90 | Sample MSR235399 (29.04.2026) | 90 gal (341 L) | LPS-only, lighter coral load | ESV B-Ionic 2-part | RM ICP-MS Assessment. Manually entered: Alk 8.0 dKH, Iodine 50.8 µg/L, Nickel 1.52 µg/L, Salinity 38.4 PSU. pH probe non-linear drift confirmed — replace probe before trusting pH readings. Per Reef Moonshiners Handbook Rev.5. Send next ICP in 2 weeks.